

”

Simulation Introduction

Unity 3D

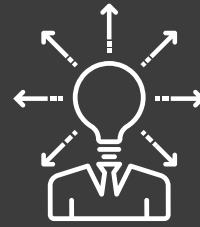
“



Demo **Autonomous Car**

The simple solution for car controlling

Simulation Concept



User Interface UI

Introduction about simulation interface

- Button
- Map



Transfer

How images are transfer



Json Command

Command to control

- Acceleration
- Steering Angle
- Brake

The Workflow of Simulation

01

Read image from camera

The builtin Camera expose the image of lane line

02

Send image interval through socketio

Interval is 0.1s per each image

03

Image processing

Real-time data and analytics provide insights for informed decision-making, leading to better outcomes.

04

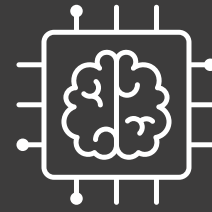
Json Command

How to compute json string to send to simulation to control the car

User Interface Unity

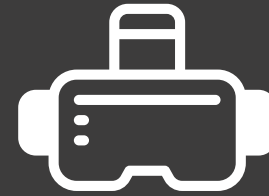


Control Button



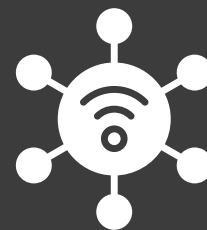
Moving Forward & Steering

W: forward
A: left
D: right
S: brake



Moving backward

Left Shift: Gear up
Left control: Gear down
To moving backward: press control for 2 times to reverse gear and press W to move backward. Shift to gear up then W to go forward

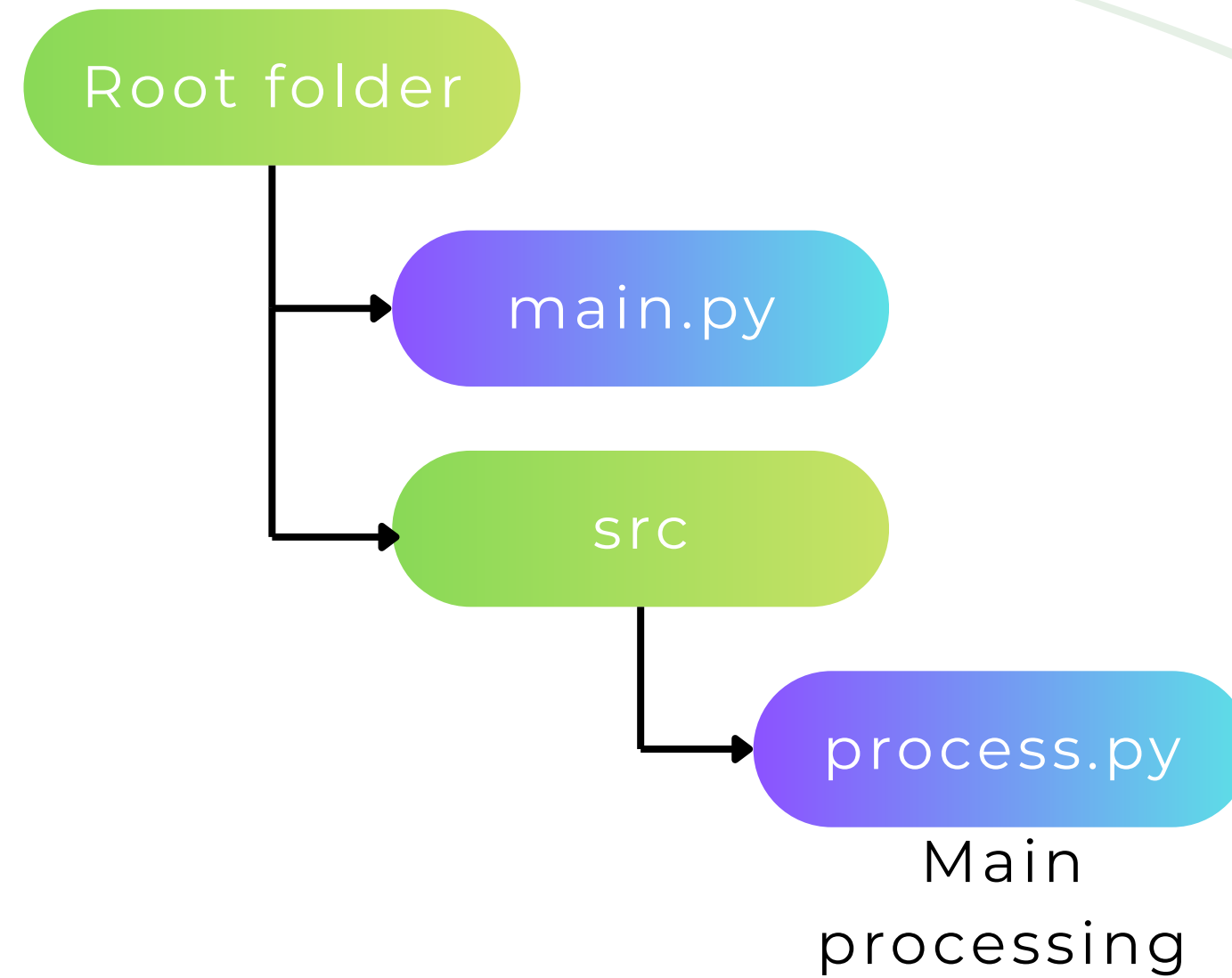


Reset & Start

R: reset
start streaming button: start streaming images

Autonomous Driving

OpenCV, AI processing



Processing part

Engine command structure:

```
{  
  "acceleration": float,  
  "steerAngle": float,  
  "brake": 0 or 1  
}
```



Get image part

- Using socketio
- Convert Base64 image to opencv format

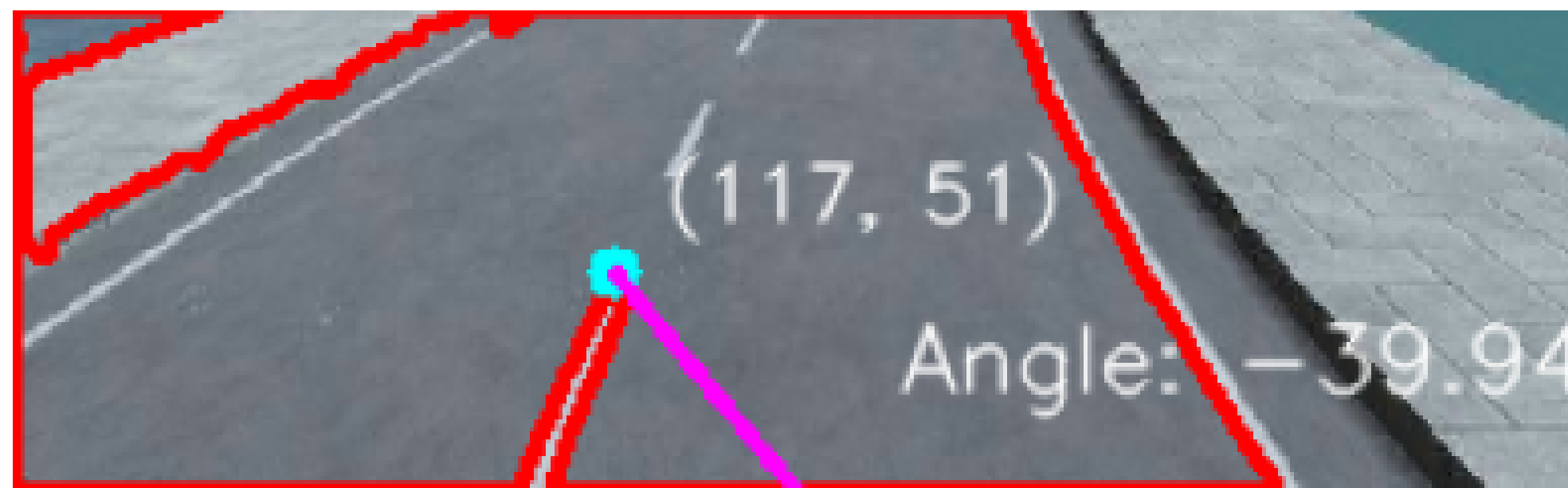


Process image and command

- # MAIN PROCESSING #
- process image to get
 - acceleration & steer angle
- process image to get traffic sign and decide next step
 - stop sign (brake car for 5s)
 - no turn right and left => then go straight

Deviation OpenCV

Largest Contour with Deviation Angle



Requirments

Mini project

- Car controlling
 - Calculate the steering angle and acceleration
 - using opencv, contour, moment
 - Calculate steering angle using Arctan
 - Suggested acceleration: 3s speed up, 1s free acceleration
- Traffic sign detection
 - Classification of stop and no turn right and left sign
 - stop sign: brake car for 5s
 - no turn right and left then go straight
- Finish the map
 - Just go until the screen show finished

”

Thank you

For joining

“